

Bill of Materials - ACT-FP-5500 'Liberty' Composite Frac Plug, 5-1/2 in.

Engineering Bill of Materials | Document BOM-FP-5500 Rev E

Product / Assembly	ACT-FP-5500
Document No. / Rev	BOM-FP-5500 Rev E
Effective Date	2025-11-20
Prepared By	A. Gutierrez
Checked By	T. Okafor
Approved By	J. Martinez, P.E.
ERP Reference	ITEM-FP-5500

Revision History

Rev	Date	Description of Change	By
A	2022-11-08	Initial release	R. Callahan
B	2023-05-16	Ceramic button count 7 to 9 per slip	A. Gutierrez
C	2024-01-24	Dissolvable Mg ball made standard	R. Callahan
D	2024-09-19	Phenolic ball option -08-P added	A. Gutierrez
E	2025-11-20	Shear screws qty 6 to 8 per ECO-2025-014; set force 55,000 lbf	R. Callahan

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Hierarchy: Level 0 = top assembly; Level 1 = component or sub-assembly; Level 2 = sub-assembly component. Indented descriptions belong to the sub-assembly immediately above. Quantities are per one (1) next-higher assembly.

Item	Lvl	Part Number	Description	Qty	UOM	Material / Spec	Notes
1	0	ACT-FP-5500	FRAC PLUG ASSY, 5-1/2 IN. COMPOSITE 'LIBERTY', 10K	1	EA	-	Rev E per ECO-2025-014
2	1	ACT-FP-5500-01	■ MANDREL, COMPOSITE, FILAMENT WOUND	1	EA	GLASS/EPOXY COMPOSITE	Millable < 12 min avg
3	1	ACT-FP-5500-02	■ ELEMENT, PACKING, SINGLE, MOLDED	1	EA	HNBR 90A (CPD ACT-H90)	Rated 300 degF
4	1	ACT-FP-5500-03	■ SLIP, UPPER, ONE-PIECE W/ BUTTONS	1	EA	CAST IRON CL 30 / CERAMIC BUTTONS	9 buttons
5	1	ACT-FP-5500-04	■ SLIP, LOWER, ONE-PIECE W/ BUTTONS	1	EA	CAST IRON CL 30 / CERAMIC BUTTONS	9 buttons
6	1	ACT-FP-5500-05	■ CONE, UPPER, COMPOSITE	1	EA	GLASS/EPOXY COMPOSITE	
7	1	ACT-FP-5500-06	■ CONE, LOWER, COMPOSITE	1	EA	GLASS/EPOXY COMPOSITE	
8	1	ACT-FP-5500-07	■ RING, LOAD, ANTI-EXTRUSION	2	EA	PHENOLIC LAMINATE	

Item	Lvl	Part Number	Description	Qty	UOM	Material / Spec	Notes
9	1	ACT-FP-5500-08	■ BALL, FRAC, DISSOLVABLE, 1.75 DIA	1	EA	MG ALLOY, DISSOLVABLE	Ships separately; -08-P phenolic optional
10	1	ACT-FP-5500-09	■ SEAT, BALL, MOLDED	1	EA	GLASS/EPOXY COMPOSITE	
11	1	ACT-FP-5500-10	■ SCREW, SHEAR, SETTING, 1/4-28 UNF	8	EA	BRASS C36000	Qty 6 -> 8 per ECO-2025-014; 55,000 lbf set force
12	1	ACT-FP-5500-11	■ GUIDE, BOTTOM, MULE SHOE	1	EA	GLASS/EPOXY COMPOSITE	
13	1	ACT-FP-AK-55	■ ADAPTER KIT, WIRELINE SETTING	1	KT	-	For ACT-ST-10 (#10) setting tool
14	2	ACT-FP-AK-55-01	■ SLEEVE, SETTING, SHORTY	1	EA	AISI 4140	
15	2	ACT-FP-AK-55-02	■ ROD, TENSION, ADAPTER	1	EA	AISI 4140 HT 28-32 HRC	

General Notes

- All elastomer components are single-use: replace at every redress from the applicable redress kit.
- Load-path metallic components require heat traceability to a mill certificate (EN 10204 Type 3.1) retained in the device history record.
- Shear devices (rings, screws, pins) are calibrated items - substitute parts are prohibited.
- Torque all threaded connections per ACT specification TS-100 (packers/plugs) or TS-200 (accessories).
- Sour service (NACE MR0175/ISO 15156) configurations use the materials noted; verify trim level on the sales order before assembly.
- Referenced documents: datasheet DS-FP-5500, applicable manual/procedure, and quality plan ACT-QP-08.

Approvals

Role	Name	Signature	Date
Design Engineering	A. Gutierrez	/signed electronically/	2025-11-20
Manufacturing Engineering	T. Okafor	/signed electronically/	2025-11-20
Quality Assurance	S. Nguyen	/signed electronically/	2025-11-20
Engineering Manager	J. Martinez, P.E.	/signed electronically/	2025-11-20